

Agriculture’s Nitrogen Solution

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Abstract

Building on Metaxoglou and Smith (2025), this paper examines how the composition of upstream agricultural land interacts with precipitation to shape downstream surface-water nitrogen concentrations. I combine 111,563 nitrogen observations from 4,094 U.S. monitoring stations from 2008 to 2023 with USDA Cropland Data Layer, PRISM climate, and HydroBASINS data to construct date-specific measures of agricultural land use and recent precipitation throughout each monitoring site’s upstream hydrological network. Two findings suggest that shifting agricultural land from annual row crops toward grass and pasture may reduce nitrogen concentrations following precipitation. First, corn and soybean acreage exhibit positive and statistically significant interactions with recent precipitation, while grass and pasture generally do not. Second, greater pasture acreage significantly attenuates the positive relationship between row-crop acreage, precipitation, and downstream nitrogen concentrations. Both relationships are concentrated within 50 km upstream of monitoring locations. Within this distance, moving from the 10th to the 90th percentile of pasture acreage reduces the estimated marginal row-crop–precipitation relationship by approximately 47 percent. Together, these results suggest that agricultural land-use composition may be an important determinant of water-quality responses to precipitation and that shifting acreage from corn and soybeans toward perennial grass and pasture may reduce nitrogen pollution associated with precipitation.

1 Introduction

Nitrogen, an essential element for the growth of flora and fauna, becomes a pollutant when present in excess, leading to significant environmental degradation. High levels of nitrogen in aquatic systems promote unchecked algal blooms, displacing aquatic life, diminishing the aesthetic and recreational value of water bodies, and posing toxic risks. In the U.S., approximately 40% of rivers and streams suffer from poor water quality due to elevated nitrogen levels (EPA, 2016). Furthermore, Dubrovsky et al. (2010) finds that nitrogen levels surpass safe drinking water thresholds in over 20% of shallow wells in agricultural zones. The nutrient overflow from the Mississippi River Basin contributes to a hypoxic zone in the Gulf of Mexico, covering 5,000 square miles, where depleted oxygen levels result in marine fatalities. The financial toll of freshwater eutrophication, primarily due to nutrient pollution, is estimated at \$2.2 billion annually (Dodds et al. 2009). Efforts to mitigate nitrogen pollution have been implemented, yet advancements in water quality have been frustratingly slow (Cheng et al. 2020) (Hansen et al. 2021).

The primary culprit for nitrogen pollution is the use of commercial fertilizers in agriculture, (Council on Environmental Quality, 2000) (David et al. 2010). Crops absorb only a portion of the available nitrogen, leaving the excess to runoff into waterways, seep into groundwater, or escape as gas through processes such as denitrification and volatilization. According to the Natural Resources Conservation Service (2017), U.S. farmland averages a nitrogen surplus of 34 pounds per acre annually, with 26 pounds of this surplus migrating into aquatic systems and the rest lost as gas. This issue is compounded by the high solubility of nitrate, a common plant-available form of nitrogen, leading models such as the Soil and Water Assessment Tool (SWAT) to predict rapid transfer of nitrogen into water bodies.

Precipitation is central to the movement of this nitrogen through agricultural landscapes. Nitrogen stored in agricultural soils and groundwater does not necessarily enter surface water contemporaneously with fertilizer application. Instead, precipitation can mobilize both recently applied and previously accumulated nutrients and transport them through the hydrological system. Recent econometric evidence demonstrates that nutrient concentrations increase following heavy precipitation and that these increases are particularly pronounced in agricultural areas (Skidmore et al., 2023). Consequently, the water-quality consequences of agricultural land use may depend not simply on how much land is devoted to agricultural production, but on how particular land uses interact with precipitation and where those land uses are located within the upstream hydrological network.

Metaxoglou and Smith (2025) provide important evidence of this relationship. Using panel data from U.S. water-quality monitoring locations, they estimate a positive relationship between upstream corn acreage and downstream nitrogen pollution. They further show that the magnitude of the relationship between corn acreage and nitrogen increases with precipitation. Their estimated contribution of an additional corn acre to downstream nitrogen is substantially smaller than estimates generated by conventional agronomic and hydrological models, a finding they interpret in the context of the growing literature on legacy nitrogen (Metaxoglou and Smith, 2025). This paper builds directly on their analysis by asking

a related question: does the composition of the upstream agricultural landscape alter the relationship between agricultural production, precipitation, and downstream nitrogen concentrations?

I focus particularly on the distinction between annual row crops and perennial grass and pasture. Corn production is especially nitrogen intensive, receiving substantially greater nitrogen fertilizer applications per acre than most other major U.S. crops. Soybeans require considerably less external nitrogen fertilizer because of biological nitrogen fixation, but corn and soybeans are commonly produced within annual row-crop rotations. Annual row-crop systems may also facilitate nutrient movement through periodic soil disturbance and periods without continuous living vegetative cover. Previous research has documented the importance of tillage, crop rotation, and land management for soil carbon and nitrogen retention (Sainju et al., 2017), while perennial grasses can maintain substantial root biomass and soil carbon and nitrogen stocks relative to annual cropping systems (Sainju et al., 2017).

Perennial grass and pasture therefore provide a potentially important contrast with annual row-crop production. Established perennial systems maintain living vegetation and root systems for a greater portion of the year, generally require less frequent soil disturbance, and can differ substantially from annual row crops in nutrient cycling and fertilizer requirements. Long-term experimental evidence also indicates that perennial grasslands can improve soil characteristics and increase soil carbon relative to annual cropping systems (Sanford et al., 2021). These characteristics provide several potential mechanisms through which perennial land cover could retain nutrients within agricultural soils or reduce their mobilization following precipitation. They motivate the central hypothesis of this paper: replacing some annual corn and soybean acreage with perennial grass or pasture may reduce the sensitivity of downstream nitrogen concentrations to precipitation.

I test this hypothesis by combining surface-water nitrogen measurements from the Water Quality Portal with annual land-use information from the USDA Cropland Data Layer, monthly precipitation and temperature data from PRISM, and the HydroBASINS hydrological network. The resulting estimation sample contains 111,563 nitrogen observations from 4,094 monitoring stations between 2008 and 2023. I link these monitoring locations to 22,071 unique Level 10 HydroBASINS sub-basins located within 300 km upstream.

The construction of the upstream network allows agricultural land use and weather conditions to be assigned according to hydrological connectivity rather than geographic proximity alone. Crop acreage is measured within hydrologically connected upstream sub-basins, and precipitation is matched to the date on which each nitrogen sample was collected. I construct precipitation measures over one-, three-, and six-month windows and divide the upstream network into mutually exclusive 50-km distance intervals. This approach allows me to examine both when precipitation matters for the relationship between agricultural land use and nitrogen and where within the upstream network that relationship is concentrated.

The empirical results provide two complementary pieces of evidence that shifting agricultural land from annual row crops toward perennial grass and pasture may reduce nitrogen concentrations following precipitation. First, corn and soybean acreage exhibit positive and statistically significant interactions with recent precipitation. The corn–precipitation inter-

action is positive over one-, three-, and six-month windows, while the soybean–precipitation interaction is positive over the one- and three-month windows. Grass and pasture acreage, in contrast, does not exhibit a statistically significant positive interaction with precipitation over the one- or three-month windows. The nitrogen response associated with precipitation therefore differs substantially between annual row crops and perennial grass and pasture.

Second, I directly test whether pasture alters the relationship between row-crop acreage, precipitation, and downstream nitrogen concentrations. The estimated crop–pasture–precipitation interaction is negative and statistically significant using both corn acreage and combined corn-and-soybean acreage. Greater pasture acreage is therefore associated with a weaker positive relationship between row-crop acreage, precipitation, and downstream nitrogen concentrations. This result is distinct from a negative direct effect of pasture on nitrogen. Rather, it indicates that the nitrogen response associated with row crops and precipitation becomes smaller as pasture acreage in the upstream agricultural landscape increases. Both relationships are strongly spatial. The positive corn–precipitation and soybean–precipitation interactions are concentrated within 50 km upstream of the monitoring location. The negative pasture-moderation relationship is also strongest within this same portion of the upstream network. In the combined row-crop specification, moving from the 10th to the 90th percentile of pasture acreage within 50 km upstream reduces the estimated marginal row-crop–precipitation relationship by approximately 47 percent. Thus, the portion of the upstream landscape in which row crops exhibit the strongest relationship with nitrogen following precipitation is also the portion in which pasture most strongly attenuates that relationship.

This paper makes three primary contributions. First, I extend Metaxoglou and Smith (2025) by moving beyond the relationship between corn acreage, precipitation, and nitrogen to examine how the composition of the agricultural landscape alters that relationship. This distinction is important in light of evidence that legacy nitrogen can remain within agricultural landscapes for decades and subsequently enter waterways following precipitation (Van Meter et al., 2017; Van Meter et al., 2018; Skidmore et al., 2023). The results suggest that the water-quality consequences of agricultural production depend not only on the acreage devoted to row crops but also on the amount of perennial agricultural land present in the same upstream landscape.

Second, I introduce a more hydrologically explicit measure of upstream agricultural exposure. Rather than assigning crop acreage to monitoring locations based solely on administrative boundaries or geographic proximity, I use Level 10 HydroBASINS to trace hydrologically connected sub-basins upstream from each monitoring location. I then decompose these exposures into mutually exclusive distance intervals. The results show that both the positive row-crop–precipitation relationship and the negative pasture-moderation relationship are concentrated within 50 km upstream.

Third, the results provide two complementary pieces of evidence consistent with the hypothesis that shifting agricultural land-use composition from corn and soybeans toward perennial grass and pasture could reduce nitrogen responses to precipitation. Row crops exhibit positive crop–precipitation interactions that are generally absent for pasture, and greater pasture acreage attenuates the estimated row-crop–precipitation relationship. This interpretation is

consistent with agronomic evidence that perennial vegetation and agricultural management can affect soil carbon and nitrogen retention (Sainju et al., 2017; Sanford et al., 2021). The empirical design does not, however, directly identify the causal effect of converting a particular acre of corn or soybeans to pasture. The results instead provide evidence consistent with land-use substitution as a potential strategy for reducing precipitation-related nitrogen pollution and motivate future work directly examining row-crop-to-pasture transitions.

The remainder of the paper proceeds as follows. Section 2 provides background on agricultural nitrogen pollution, legacy nitrogen, the economic damages associated with nitrogen pollution, and perennial grass and pasture as a potential mitigation strategy. Section 3 describes the water-quality, agricultural land-use, climate, and hydrological data and the construction of upstream exposures. Section 4 presents the empirical approach. Section 5 reports the results, including the spatial decomposition and pasture-moderation estimates. Section 6 concludes and discusses implications for agricultural land use and water-quality policy.

2 Background

2.1 Nitrogen fertilizer and water pollution

Nitrogen, while being the predominant element in the atmosphere, is vital for plant growth and agricultural productivity. Yet, for atmospheric nitrogen to be utilized by plants, it must first be transformed into accessible forms such as ammonium or nitrate through a process known as fixation. Biological fixation alone falls short in providing sufficient nitrogen for optimal crop yields. The development of the Haber-Bosch process in the early 20th century marked a significant advancement, synthesizing ammonia from atmospheric nitrogen and hydrogen under intense heat and pressure, thereby helping to dramatically boosting agricultural output post-World War Two (Capel et al., 2018, among others). Furthermore, the nitrogen cycle includes processes like mineralization and nitrification, which respectively convert organic nitrogen from sources like manure and crop residues into ammonia, and then ammonia into nitrate, increasing the availability of nitrogen to plants.

Several processes within the nitrogen cycle diminish the availability of nitrogen to plants, namely denitrification, volatilization, immobilization, and leaching. Denitrification transforms nitrate into gaseous forms like nitrous oxide—a known greenhouse gas—thus removing it from the soil. Volatilization causes nitrogen to escape as ammonia gas into the atmosphere. Immobilization, which can be thought of as the opposite of mineralization, involves soil organisms absorbing plant-available nitrogen, making it inaccessible to crops. Leaching is particularly problematic because soil particles do not effectively hold onto nitrates, allowing them to move freely with water through the soil.

In the U.S. corn receives significantly more nitrogen fertilizer than other major crops. This is evident in the surveys conducted by the USDA’s Agricultural Chemical Use Program from 2021 to 2023. In the 2021 survey 12.3 billion pounds of nitrogen were applied to corn at

an average rate of 150 pounds per acre, substantially more than the 695 million pounds used on cotton at an average rate of 95 pounds per acre in that same year. The nitrogen figures also far surpass the figures associated with wheat and soy from the 2022 and 2023 surveys respectively with 2.7 billion pounds of nitrogen applied to wheat at 72 to 82 pounds per acre, and 537 million pounds of nitrogen applied to soy at an average of 22 pounds per acre. These four crops account for roughly two-thirds of the country’s principal crop acres. Soybeans, which are legumes that fix atmospheric nitrogen to the soil, require minimal nitrogen fertilizer. Notably, there is a belief that farmers apply too much nitrogen in general, a practice often termed “insurance” nitrogen, due to fear of applying too little nitrogen, and lack of immediate consequences of applying too much. (Mitsch et al., 1999); CENR, 2000).

According to the NRCS’s APEX simulation model results for 2003 to 2006 29% of nitrogen applied as chemical fertilizer or manure to cultivated cropland acres is not taken up by the crops (NRCS, 2017). This model also produced a mean of the average annual estimates of total-edge-of field surplus nitrogen of 34 lbs per acre per year. Depending on the type of fertilizer the amount of surplus nitrogen varied significantly, with the highest estimate of 56 lbs per acre per year for cultivated cropland acres receiving manure. The NRCS’s APEX model estimates that 26 lbs of the 34 lbs in average losses enters waterways, while 6.5 lbs is volatilized and 2.4 lbs is identified. Hendricks et al (2014) used standard agronomic and hydrologic models, APEX and SWAT, and reached a similar estimate of 34.7 lbs per corn acre per year.

Surplus nitrogen running off into waterways can lead to eutrophication, which is the massive overgrowth of algae, where this growth of algae is referred to in the literature as an algal bloom. Algal blooms can be harmful to the local ecology and disrupt native populations of aquatic flora and fauna by disrupting access to sunlight, increasing turbidity, and even creating hypoxic environments as they decompose. Harmful algal blooms, a specific type of algal bloom which emits toxins into the water, can cause adverse health effects in humans who consume shellfish which have been contaminated by being in the same waters as these toxins. Apart from the negative impacts caused by the algal blooms, excess nitrogen in drinking water can cause “blue baby” syndrome for infants younger than six months. “Blue baby” syndrome currently is not a common issue, as there hasn’t been a case confirmed since 2000, but an increase in the leaching of nitrogen into groundwater may make syndrome more common.

The Gulf hypoxic zone was once the second largest hypoxic zone in the world, with its maximum area measured at 8,800 square miles in 2017, while it has been most recently measured at 6,705 square miles in 2024. The Mississippi River Basin’s outflow of nitrogen to the Louisiana Gulf has caused the growth of this hypoxic zone which was first measured at 4000 square miles in 1985. The 2024 area figure is well above the EPA’s Hypoxia Task Force goal of 1931 square miles and is higher than the 5 year average area of 4298, indicating the negative affect of nitrogen runoff from the MRB is still impacting coastal resources and habitats in the Gulf. Fertilizers have been estimated to have contributed to 50% (CENR(2000)) and 76% (David et al. (2010)) of the total nitrogen riverine export from the MRB to the Gulf.

Besides nitrogen fertilizer, other sources of nutrient pollution in streams and rivers can come

from dairy, cattle, hog, poultry, and aquaculture operations. The wastewater from these sources is often untreated and contains high levels of nitrogen, typically as ammonia, though nitrate levels can also be elevated. Urban and suburban environments contribute substantial runoff from surfaces like lawns, parking areas, rooftops, and roads. Key point sources for nutrient discharge directly into water bodies include municipal wastewater treatment facilities. Additionally, the burning of fossil fuels in vehicles and power stations adds to the levels of airborne nitrates, which can return back to surface through wet deposition from precipitation or as gases and particles in dry deposition and then eventually runoff into waterways.

2.2 Damages from nitrogen pollution

The true cost of nitrogen pollution is difficult to estimate since there are various pathways associated with damages to water quality and the degree in which these damages occur varying over space and time (Gourevitch et al. (2018)). This in turn makes it challenging to construct a single cost metric on society for nitrogen pollution into waterways. Keeler et al. (2012) present a valuation template that uses biophysical and economic models that links nitrogen activities to changes in value of goods and services, considering several sources of goods and services. Using this valuation template Taylore and Heal (2021), excluding health effects, estimate the effects of algal blooms that resulted from surplus nitrogen to have an external cost of \$583 per ton of fertilizer applied. Furthermore, they estimate damages to coastal counties downstream of nitrogen fertilization from \$370-\$1,400 per ton.

Studies also incorporate damages outside of local water quality impacts such as Sobota et al (2015). They estimate damages for air, climate, land, freshwater, drinking water and coastal zones. Sobata et al. (2015) incorporates estimates from Compton et al. (2011) and Van Grinsven et al. (2013a) to estimate damages, in terms of dollars per kg of excess nitrogen, from costal nitrogen lading due to recreational use of \$6.38, declines in fisheries and marine habitat of \$15.84, and increased eutrophication in freshwaters of \$16.10. Van Grinsven et al. (2013b) investigate damages from nitrogen pollution in terms of human health, ecosystems and climate in the E.U in 2008. They estimate a total cost of nitrogen loss from croplands in the range of 25-100 billion euros a year (\$37.5-\$150 billion a year), implying a cost of 4.11-16.43 euros per lb of nitrogen.

These aren't the only studies that have estimated the costs to society associated with nitrogen pollution. UCS (2020) estimates that excess nitrogen from Midwest cropland causes between \$552 million and \$2.4 billion in damages to fisheries and marine habitat in the Gulf. Ho et al. (2019) estimates losses of more than \$4 billion annually in the U.S from freshwater algal blooms alone. Dodds et al. (2009) estimate potential annual value losses in recreational water usage, real estate, recovery spending and drinking water in the U.S to combine to \$2.2 billion annually. Anderson et al (2000) looks at harmful algal blooms in the U.S from 1987-1992 and their effect on public health, commercial fishery, recreation and monitoring management. They estimate a 15-year capitalized impacts at "low", "average" and "high" levels which are: \$309 million, \$449 million and \$743 million respectively.

2.3 An Agricultural Solution

It has been well documented that corn acreage along with other commodity row crops acreage where nitrogen fertilizers are applied are a leading source of nitrogen deposition into surface waters in the U.S. ((Dodds et al, 2016) (EPA, 2022) ((Paudel and Crago, 2020). There are three underlying mechanisms a part of conventional row crop cultivation and harvesting that lends itself to the creation of nitrogen runoff: the mass application of nitrogen fertilizers, significant soil disturbance from tilling and the subsequent lack of soil coverage in and out of growing season. In a well-managed pasture-based livestock operation all three of these mechanisms are significantly less severe, or completely non-existent.

First off, well-managed pasturelands require no to very little amounts of added fertility due to livestock directly depositing nutrients via manure and the common practice of planting at least one kind of legume, such as a white clover, in with the grass mix to help fix atmospheric nitrogen to the soil. Secondly, even if a pasture is not well managed, we see no frequent mechanical disturbance of soil through tilling or plowing, with very rare cases of initial plowing or tilling when first establishing a pasture. Lastly, in a pasture system we almost never see baren soil, in or out of growing season, since grass is quite literally a “cover crop”, and is not planted in rows with baren soil in between each row like corn, soy and cotton. Plus, even after grass has been harvested by ruminant livestock, the bottom 1-3 inches are always preserved due to the eating habits of cattle, ewes and goats, meaning the act of “harvesting” does not leave the soil baren (Undersander et al, 2002). In contrast to commodity row crops, the act of planting, maintaining and harvesting in a pasture-based operation does not cause significant soil degradation and subsequent nutrient runoff.

In addition to preventing nutrient runoff, pasturelands can be used as a tool to help improve the environment. Perennial, diverse grass mixes can actually improve soil health and provide more sustainable yields than monoculture grains (Sanford et al, 2021). Another finding from the Sanford et al 26 year research trial, is that all but one kind cropping method analyzed, even the methods with cover crops, saw carbon losses in the top soil. The cropping methods that gained carbon in its topsoil were perennial grasslands where dairy heifers directly grazed. This finding provides evidence for another environmental advantage perennial pasturelands have over commodity row crops, the ability to store atmospheric carbon in the soil known as a “carbon sink”. It is estimated that pasture soils account for approximately 30% of total global soil organic carbon (Silveira et al. , 2022).

For grazing ruminant animals on perennial pastureland to be a true “agricultural solution” it must not only positively impact the environment, but it must also be economically feasible. The three main costs of raising any animal are feed, labor and capital costs; for ruminant animals, pasture-based operations have lower costs across all three compared to confinement operations. This is highlighted in a four year study of Jersey and Holstein cows, where on average pasture-fed cows were 11.1% less productive on average than confinement raised cows, but labor, capital, and feed savings made it such that pasture-based operations were financially competitive with confinement operations (White et al, 2002)

3 Data sources and descriptive analysis

In this section I discuss the main data sources used in the analysis. I then discuss trends in the spatial variation of nitrogen pollution over time and conclude with summary statistics on the variables of interest.

3.1 Data sources

Water quality. The data on nitrogen concentrations are from the Water Quality Portal (WQP), which is a service sponsored by the USGS, the EPA and the National Water Quality Monitoring Council. In total, the WQP contains data collected by over 400 state, federal, tribal and local agencies with more than 300 million water quality records. I accessed the data described as “total nitrogen [nitrate+nitrite+ammonia+organic N], water, unfiltered, milligrams per liter” using parameter code 00600. For the following analysis, I measure nitrogen pollution in milligrams per liter, and I limit the data to only routine samples of surface waters.

Following Metaxoglou and Smith, the monitoring sites used in the main analysis are located east of the 100th meridian and outside of Florida. With the following restrictions on the data, the sample held 111,563 observations from 4,094 USGS monitoring stations between 2008 and 2023. These monitoring locations are hydrologically linked to 22,071 unique Level 10 HydroBASINS sub-basins within 300 km upstream. Figure 1 shows the spatial distribution of the USGS monitoring stations in the sample as red dots.

Hydrological-Basins. HydroBASINS from HydroSHEDS represents a series of vectorized polygon layers that depict sub-basin boundaries of watersheds at a global scale. HydroBASINS has been extracted from the gridded HydroSHEDS core layers at 15 arc-second resolution. The two key features of the HydroBASINS are the upstream-downstream connectivity between basins, and the Pfafstetter coding system that has been implemented in the HydroBASINS product offering 12 hierarchical nested sub-basin breakdowns globally. At the highest level of sub-basin breakdown there are one million individual sub-basins globally with an average area of 130.6 km², due to computational restrictions, the analyses will be evaluated at the 10th level. This part of the data, along with the crop data, is one of the more significant deviations from Metaxoglou and Smith as they used the Hydrological Flows map to trace out distance upstream, while I use the upstream-downstream relationship already embedded in the HydroBASINS data. Figure 2 shows the 22,071 of the 10th level in the sample. The hydro-basins exclude Florida, and extend out to a latitude of -105 for redundancy. Figure 1 displays the upstream hydro-basins in gray used in the analysis.

Crops. Annual satellite data on planted crop acres is available from the NASS Crop Data Layer Program. Starting in 2008, the program has national data available to download, where the data is broken down into 30 meters by 30 meters cells. I aggregate the acreage of soy, corn and grassland/pasture for each year available within the hydro-basins in the sample. I focus on these three crops over the hydro-basins in the sample for two reasons:

(1) they are all primarily rainfed in this region of the U.S (2) from 2008 to 2023 over 95% of the total corn and soy acreage in the U.S was found east of the 100th meridian and outside of Florida. Figures 2,3 and 4 show the 2023 spatial distribution for corn, soy and grass/pastureland in the sample. Where each dot represents a 30m x 30m square where corn, soy or grass/pastureland was planted in 2023.

Weather. Weather data come from the PRISM Climate Group. Precipitation is measured monthly for each Level 10 HydroBASINS sub-basin and is linked to individual water-quality observations according to the date on which the nitrogen sample was collected.

The primary precipitation measures are constructed over one-, three-, and six-month windows ending in the observation month. The one-month measure captures precipitation during the calendar month in which the nitrogen sample was collected. The three-month measure averages precipitation over the observation month and preceding two months, while the six-month measure averages precipitation over the observation month and preceding five months. These alternative windows allow the empirical analysis to distinguish relationships associated with relatively recent precipitation from those associated with precipitation accumulated over longer periods.

For each monitoring location, monthly precipitation is aggregated across hydrologically connected upstream sub-basins using sub-basin area weights. I construct precipitation measures for the complete network within 300 km and separately for each mutually exclusive 50-km distance interval. Consequently, precipitation and crop acreage in the spatial specifications refer to the same portion of the upstream hydrological network.

The regressions additionally control for broader temperature conditions using annual mean temperature, January minimum temperature, and August maximum temperature measured over the upstream area within 300 km. These variables account for temperature conditions that may influence crop production, biological nitrogen processes, and measured surface-water nitrogen concentrations.

3.2 Descriptive analysis

Metaxoglou and Smith describe the historical trend of nitrogen pollution using the Water Quality Portal data from 1970-2017 and historical county-level fertilizer data. Their main descriptive analysis showed that average nitrogen concentration peaked at 3mg/L in 1982 and declined to and remained at an average of 2.3 mg/L since the 2000s. Nitrogen fertilizer use peaked at 12.5 million metric tons in 2013 before declining to 11.8 million metric tons in 2015.

Table 1 reports descriptive statistics for the principal water-quality, agricultural land-use, and weather variables used in the analysis. The sample contains 115,083 nitrogen observations collected between 2008 and 2023. Nitrogen concentrations average 3.00 mg/L, with a median of 1.50 mg/L and an interquartile range of 0.78 to 3.60 mg/L. The difference between the mean and median reflects the right-skewed distribution of observed nitrogen concentrations.

Panel A summarizes agricultural land use and weather throughout the upstream hydrological network within 300 km of each monitoring location. Mean upstream corn and soybean acreage are approximately 131,000 and 127,000 acres, respectively, producing an average of approximately 259,000 acres of row-crop production. Mean grass and pasture acreage is approximately 147,000 acres. These land-use measures are highly heterogeneous across monitoring locations. Median row-crop acreage is only approximately 10,000 acres, compared with a mean of approximately 259,000 acres, reflecting substantial variation in the agricultural intensity and size of the upstream areas linked to monitoring locations.

Recent precipitation also varies substantially across observations. Mean precipitation during the month of sampling is approximately 104 mm, with an interquartile range of approximately 63 to 134 mm. Average precipitation over the preceding three- and six-month windows is approximately 100 and 97 mm per month, respectively. As expected, variation decreases as the precipitation window becomes longer.

Panel B focuses on the 0–50 km upstream area, which is particularly important for the spatial specifications developed below. Monitoring observations are associated with an average of approximately 36,500 acres of row crops and 19,400 acres of grass and pasture within this area. The corresponding medians are approximately 7,500 and 9,500 acres, again indicating considerable right-skewness in upstream agricultural acreage. This variation in nearby crop and pasture acreage, together with variation in recent precipitation, provides the spatial and temporal variation used to examine whether the relationship between agricultural production and downstream nitrogen concentrations depends on both the composition and location of the upstream agricultural landscape.

4 Empirical Approach

The empirical analysis examines whether the relationship between upstream agricultural land use and downstream surface-water nitrogen concentrations depends on recent precipitation and whether that relationship varies with the composition of the upstream agricultural landscape. The analysis proceeds in four steps. I first estimate the interaction between crop acreage and precipitation using cumulative upstream exposure within 300 km. I then decompose this relationship by hydrological distance. Next, I test whether grass and pasture acreage moderates the relationship between row-crop acreage and precipitation. Finally, I use the estimated interaction coefficients to calculate the crop–precipitation relationship across the observed distribution of grass and pasture acreage.

All specifications include monitoring-site and observation-year fixed effects. Monitoring-site fixed effects absorb time-invariant differences across sampling locations, including persistent differences in watershed characteristics, topography, soils, and other fixed characteristics of the surrounding landscape. Observation-year fixed effects absorb common annual shocks affecting nitrogen concentrations across monitoring locations. Standard errors are clustered by monitoring site to allow for arbitrary serial correlation and heteroskedasticity within locations over time.

4.1 Framework

I adopt an identical framework as Metaxoglou and Smith where the average nitrogen concentration at monitoring site i in year t is y_{it} and it is affected by upstream fertilizer applications. Farmers choose annually which crop to plant. Conditional on crop choice, they decide how much nitrogen fertilizer to apply and when to apply it. Some of this nitrogen reaches site i , potentially with a (long) delay. Let the high-dimensional vector a_{it} contain information on which crop was planted on each field upstream from site i in year t and prior years. Similarly, the vector p_{it} denotes precipitation in each field upstream from site i in year t and prior years. I include precipitation because it plays an important role in transporting nitrogen into waterways. The average nitrogen concentration at site i in year t is described by the following equation:

$$y_{it} = f_i(a_{it}, p_{it}, z_{it}, \xi_{it}), \quad (1)$$

where z_{it} is a vector of other variables such as weather, fertilizer prices, other sources of nitrogen, and land management practices, and ξ_{it} denotes unobservables. Acreage affects concentration through the application of fertilizer on crop choice.

For the same reasons as Metaxoglou and Smith I focus on the reduced-form effect of crop acreage on nitrogen concentration. First off, crop acreage is much more precisely measured than nitrogen fertilizer application, the data I have is using a 30 meter by 30 meter cell resolution compared to total nitrogen sales at the county level. Furthermore, the concentration of nitrogen that ends up in surface waters does not come from the amount of nitrogen applied, but rather the amount of nitrogen not taken up by the crops it was applied to. Finally, agricultural policies tend to focus on changes in land use to mitigate nitrogen pollution, rather than monitoring nitrogen use.

The analysis will focus on average effects rather than the marginal effect of crop acreage. This is because the marginal effect of crop acreage in one location on nitrogen concentrations depends on the characteristics of the landscape around and upstream of the monitoring site such as the type of waterways, soil types, land slope etc... I make the same assumption that the marginal effect of crop acreage on concentration depends on distance, but I have a different approach to measuring and using distance in analysis. Differing from Metaxoglou and Smith, I measure distance in kilometers from the centroid of the hydro-basin in which the nitrogen sample was taken to the centroids of the hydro-basins that flow into the original hydro-basin. As our main explanatory variables, I use total crop acreage of all the hydro-basins upstream of the monitoring site within several distance bins. I explore the effects using distance bins ranging from 50 kilometers to 300 kilometers.

4.2 Crop Acreage and Precipitation

I begin by estimating whether the relationship between upstream crop acreage and nitrogen concentration varies with recent precipitation. For nitrogen observation s collected at

monitoring site i in year t , I estimate

$$N_{ist} = \beta_1 Crop_{ist} + \beta_2 P_{ist}^k + \beta_3 (P_{ist}^k)^2 + \beta_4 (Crop_{ist} \times P_{ist}^k) + \mathbf{W}_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{ist}, \quad (2)$$

where N_{ist} is the measured nitrogen concentration in mg/L. $Crop_{ist}$ denotes upstream crop acreage within 300 km of monitoring site i , expressed in millions of acres. I estimate Equation 2 separately using corn, soybean, and grass/pasture acreage.

P_{ist}^k denotes mean upstream precipitation over window k , where $k \in \{1, 3, 6\}$ months. The precipitation window ends in the month in which the nitrogen observation is collected. I include precipitation quadratically to allow the relationship between precipitation and nitrogen concentration to be nonlinear. The vector $\mathbf{W}_{it}$ contains annual mean temperature, January minimum temperature, and August maximum temperature. Outside of precipitation weather and climate are important for the analysis. The more conducive the temperature is for crop growth, the more likely the crops are to take up the nitrogen and not have it wash away. The terms α_i and λ_t denote monitoring-site and observation-year fixed effects, respectively.

The coefficient of primary interest is β_4 . Rather than imposing a constant relationship between crop acreage and nitrogen concentration, Equation 2 allows that relationship to vary with recent precipitation. This is motivated by the role of precipitation as the main source of crop irrigation and the main pathway of nitrogen to converge down watersheds to surface waters. A positive β_4 indicates that the relationship between upstream crop acreage and measured downstream nitrogen concentration becomes more positive as recent precipitation increases.

The crop main effect, β_1 , should therefore not be interpreted independently of the interaction. The marginal relationship between crop acreage and nitrogen concentration at a given precipitation level is

$$\frac{\partial N_{ist}}{\partial Crop_{ist}} = \beta_1 + \beta_4 P_{ist}^k. \quad (3)$$

Estimating the model using alternative precipitation windows allows me to examine whether the crop-precipitation relationship is associated primarily with recent weather or persists over longer periods.

4.3 Spatial Decomposition of Upstream Exposure

The cumulative 300-km specification treats agricultural acreage throughout the upstream network as a single exposure. Nitrogen transport, however, may depend on hydrological distance. I therefore estimate a second specification that decomposes upstream crop acreage and precipitation into six mutually exclusive distance intervals.

Let d index the upstream distance intervals 0–50, 50–100, 100–150, 150–200, 200–250, and 250–300 km. I estimate

$$N_{ist} = \sum_d \left[\beta_d Crop_{istd} + \gamma_d P_{istd} + \delta_d P_{istd}^2 + \theta_d (Crop_{istd} \times P_{istd}) \right] + \mathbf{W}'_{it} \phi + \alpha_i + \lambda_t + \varepsilon_{ist}. \quad (4)$$

$Crop_{istd}$ measures crop acreage located in distance interval d , while P_{istd} measures the one month precipitation over the same portion of the upstream hydrological network. Because all six distance intervals enter simultaneously, θ_d captures the crop–precipitation relationship associated with acreage in a particular upstream interval conditional on exposure in the remaining intervals.

The main spatial analysis uses the one-month precipitation window. This choice follows from the cumulative estimates and allows the distance decomposition to focus on the short-run relationship between recent precipitation and measured nitrogen concentrations. I estimate Equation 4 separately for corn and soybean acreage.

The distance decomposition is not interpreted as measuring the travel time of an individual unit of nitrogen. Instead, it provides a flexible way of identifying whether the empirical relationship between agricultural land use, precipitation, and nitrogen concentration is concentrated relatively near the monitoring location or distributed more broadly throughout the upstream network.

4.4 Pasture Moderation

The previous specifications establish the relationship between row-crop acreage, precipitation, and nitrogen concentration but treat grass and pasture acreage as an independent land-use category. The central extension of the analysis asks whether the composition of the upstream agricultural landscape alters the precipitation-dependent relationship between row crops and nitrogen concentration.

I estimate the following specification:

$$\begin{aligned} N_{ist} = & \beta_1 Crop_{ist} + \beta_2 Grass_{ist} + \beta_3 P_{ist} + \beta_4 P_{ist}^2 \\ & + \beta_5 (Crop_{ist} \times P_{ist}) + \beta_6 (Grass_{ist} \times P_{ist}) \\ & + \beta_7 (Crop_{ist} \times Grass_{ist}) \\ & + \beta_8 (Crop_{ist} \times Grass_{ist} \times P_{ist}) + \mathbf{W}'_{it} \gamma + \alpha_i + \lambda_t + \varepsilon_{ist}. \end{aligned} \quad (5)$$

Here, $Grass_{ist}$ denotes upstream grass and pasture acreage and P_{ist} is one-month precipitation. I estimate Equation 5 first using corn acreage and then using a broader row-crop

measure equal to the sum of corn and soybean acreage, which I refer to as "row crop" acreage in the results tables. All constituent and lower-order interaction terms are included so that the three-way interaction can be interpreted correctly.

The coefficient of primary interest is β_8 . A negative estimate of β_8 indicates that the positive crop–precipitation relationship becomes smaller as grass and pasture acreage increases. This is conceptually different from a negative direct effect of grass or pasture on nitrogen concentration. The specification tests whether perennial grass and pasture land cover *moderates* the relationship between row-crop acreage and nitrogen concentrations following precipitation.

I additionally estimate a spatial version of Equation 5. Crop acreage, grass/pasture acreage, and precipitation are divided into the same six mutually exclusive 50-km upstream distance intervals used in Equation 4, and the complete interaction structure is estimated simultaneously across the six rings. This specification identifies whether pasture moderation is more effective relatively close to the monitoring site.

4.5 Marginal Crop–Precipitation Relationships

The three-way interaction coefficient in Equation 5 does not by itself describe the magnitude of pasture moderation. I therefore use the estimated coefficients to calculate the crop–precipitation relationship at different levels of upstream grass and pasture acreage.

Differentiating Equation 5 with respect to crop acreage and precipitation gives

$$\frac{\partial^2 N_{ist}}{\partial Crop_{ist} \partial P_{ist}} = \beta_5 + \beta_8 Grass_{ist}. \quad (6)$$

Equation 6 measures how the relationship between crop acreage and nitrogen concentration changes with precipitation at a given level of grass and pasture acreage. When $\beta_5 > 0$ and $\beta_8 < 0$, the estimated crop–precipitation relationship declines as grass and pasture acreage increases.

I calculate this quantity across the observed distribution of grass and pasture acreage and construct 95-percent confidence intervals using the full clustered variance–covariance matrix of the estimated coefficients. The primary marginal-effect analysis focuses on the 0–50 km upstream interval because the spatial regressions indicate that both the crop–precipitation relationship and pasture moderation are most pronounced within this portion of the upstream network. I report results using combined corn and soybean acreage as the primary row-crop measure and use the corn-only specification as a robustness check.

5 Results

5.1 Crop Acreage, Precipitation, and Nitrogen Concentrations

Table 2 presents estimates of the relationship between upstream agricultural land use, recent precipitation, and downstream nitrogen concentrations. The specifications separately consider corn, soybean, and grass/pasture acreage within 300 km upstream of each monitoring station and allow the relationship between each land-use measure and nitrogen concentration to vary with precipitation over the preceding one-, three-, and six-month windows. All specifications include monitoring-station and observation-year fixed effects as well as controls for annual mean temperature, January minimum temperature, and August maximum temperature.

Across the three precipitation windows, the estimated main effects of corn acreage alone are not statistically distinguishable from zero. As expected, the interaction between corn acreage and precipitation is positive and statistically significant in each specification. The estimated interaction coefficients are 0.0048 for precipitation during the month of sampling, 0.0066 for average precipitation during the preceding three months, and 0.0052 for average precipitation during the preceding six months. The one- and three-month coefficients are statistically significant at the one-percent level, while the six-month estimate is also positive and statistically significant. These results indicate that the relationship between upstream corn acreage and downstream nitrogen concentrations becomes more positive as recent precipitation increases. This comes as no surprise as it is well documented in biological sciences that corn production causes nitrogen runoff, and in the sample precipitation is the main form of irrigation.

The soybean specifications exhibit a similar pattern over shorter precipitation windows. The interaction between soybean acreage and precipitation is positive and statistically significant for the one-month and three-month windows, with estimated coefficients of 0.0054 and 0.0062, respectively. The corresponding six-month interaction coefficient is smaller and is not statistically distinguishable from zero. This result is more novel than the corn result. Soybean is a nitrogen fixing legume and production requires no external nitrogen fertilizer to be applied, so it is less obvious that precipitated soy acreage causes nitrogen runoff. The analysis is likely picking up the excess nitrogen from corn production that has been rotated into soy production combined with the common soil degrading production technologies of both soy and corn. Taken together, the corn and soybean estimates suggest that the relationship between row-crop acreage and downstream nitrogen concentrations is particularly associated with relatively recent precipitation rather than precipitation measured over longer periods.

The grass/pasture specifications differ from those for corn and soybeans. The interaction between grass/pasture acreage and precipitation is not statistically significant using either the one-month or three-month precipitation measure. The interaction only becomes positive and statistically significant at the five percent level in the most coarse six-month specification. Consequently, these baseline regressions provide little evidence that pasture acreage has a

strong positive or negative relationship with nitrogen concentrations following precipitation. Instead, the results motivate examining whether pasture affects the relationship between row-crop acreage and precipitation, rather than treating pasture as an independent substitute for row crops.

Overall, the cumulative specifications establish two features of the data. First, the relationship between upstream row-crop acreage and downstream nitrogen concentrations depends importantly on recent precipitation. Second, the strongest and most consistent relationships occur over relatively short precipitation windows, in this case the one month window. For this reason, the spatial and pasture-moderation specifications that follow focus primarily on precipitation during the month of the nitrogen measurement.

5.2 Spatial Distribution of the Crop–Precipitation Relationship

The cumulative 300-km specifications identify the overall relationship between upstream agricultural acreage and nitrogen concentrations but do not identify where within the upstream network that relationship originates or is strongest. To examine its spatial distribution, Table 3 replaces the cumulative acreage and precipitation measures with mutually exclusive upstream distance rings of 0–50, 50–100, 100–150, 150–200, 200–250, and 250–300 km. Because the previous results indicate that the crop–precipitation relationship is strongest over short precipitation windows, these specifications use precipitation during the month of sampling.

The results reveal substantial spatial concentration in the closest distance bin for the estimated crop–precipitation relationship. For corn acreage, the interaction between corn and precipitation within 0–50 km upstream is 0.1229 and is statistically significant at the one-percent level. In contrast, the corresponding interaction coefficients for each of the five distance rings beyond 50 km are substantially smaller and are not statistically distinguishable from zero.

The soybean estimates produce a similar spatial pattern. The soybean–precipitation interaction within the first 50 km upstream is 0.0703 and is statistically significant at the one-percent level. As with corn, none of the estimated soybean–precipitation interactions in the distance rings beyond 50 km are statistically significant.

These results help clarify the cumulative estimates in Table 2. Although the cumulative measure incorporates agricultural acreage throughout the 300-km upstream network, the exclusive-ring estimates indicate that the positive relationship between row-crop acreage, recent precipitation, and downstream nitrogen concentration is most concentrated in the portion of the upstream network nearest the monitoring station. This pattern is consistent across both corn and soybean acreage.

The spatial concentration of the estimates is also important for evaluating the role of other agricultural land uses. If pasture alters the transmission of nitrogen from row-crop landscapes following precipitation, the preceding results suggest that such moderation should be most apparent within the relatively proximate upstream landscape where the crop–precipitation

relationship itself is strongest.

5.3 Pasture as a Moderator of the Crop–Precipitation Relationship

Table 4 examines whether the relationship between row-crop acreage, recent precipitation, and downstream nitrogen concentrations varies with the amount of grass/pasture acreage in the upstream landscape. The specifications introduce a triple interaction between crop acreage, grass/pasture acreage, and precipitation. The first specification uses corn acreage as the crop measure, while the second combines corn and soybean acreage into a broader measure of row-crop production. Both specifications use the one-month precipitation measure and cumulative agricultural exposure within 300 km upstream.

For corn, the estimated interaction between corn acreage and precipitation is positive and statistically significant, with a coefficient of 0.0101. The coefficient on the triple interaction between corn acreage, grass/pasture acreage, and precipitation is -0.0048 and is statistically significant at the one-percent level. Thus, as grass/pasture acreage increases, the estimated relationship between corn acreage and nitrogen concentrations following precipitation becomes smaller. This provides preliminary evidence of the main novel contribution of the paper: perennial pasture and grassland causing a decrease in nitrogen runoff from corn acreage.

The broader row-crop specification produces the same qualitative result. The interaction between row-crop acreage and precipitation is positive and statistically significant, with an estimated coefficient of 0.0049. The row-crop–pasture–precipitation interaction is -0.0021 and is statistically significant. The similarity of the results using corn alone and combined corn-and-soybean acreage indicates that the moderating relationship is not specific to the definition of corn acreage but to industrialized row crop acreage as a whole.

Importantly, these estimates should not be interpreted as evidence that pasture acreage reduces nitrogen concentrations on its own. The pasture main effect and the pasture–precipitation interaction are not statistically distinguishable from zero in these specifications. Rather, the negative triple-interaction coefficients indicate that the relationship between precipitation caused nitrogen runoff from row-crop acreage weaker as pasture acreage in the upstream landscape increases. The relevant empirical result is therefore one of moderation: landscapes containing more pasture exhibit a weaker positive crop–precipitation relationship than otherwise similar landscapes containing less pasture.

This distinction changes the interpretation of the role of pasture relative to a simple comparison of nitrogen concentrations across land uses. The estimates suggest that agricultural land-use composition matters for the relationship between precipitation and downstream nitrogen concentrations. In particular, the association between row-crop acreage and nitrogen concentrations following precipitation depends on the amount of pasture present elsewhere in the upstream landscape.

5.4 Spatial Heterogeneity in Pasture Moderation

Table 5 decomposes the pasture-moderation relationship across the same mutually exclusive upstream distance rings used in Table 3. This specification allows the interaction between crop acreage, pasture acreage, and one-month precipitation to vary across the upstream network.

To little surprise, the results indicate that the negative pasture-moderation relationship is strongest within the first 50 km upstream. In the corn specification, the estimated corn–pasture–precipitation interaction for the 0–50 km ring is -1.767 and is statistically significant at the one-percent level. None of the corresponding triple interactions for the five distance rings beyond 50 km are statistically distinguishable from zero.

The row-crop specification again produces a similar pattern. The estimated row-crop–pasture–precipitation interaction within 0–50 km is -0.5998 and is statistically significant at the one-percent level. The coefficient for the 50–100 km ring is positive, at 0.1065 , and statistically significant at the 5 percent level, while the estimated interactions for the remaining distance rings are not statistically significant. The estimates therefore do not imply that pasture has the same moderating relationship at every point in the upstream network. Rather, the largest negative estimate is concentrated within the first 50 km upstream.

This spatial pattern closely corresponds to the results in Table 3. The strongest positive crop–precipitation relationships occur within 50 km upstream, and the strongest negative crop–pasture–precipitation interactions occur within this same portion of the upstream network. Together, these findings suggest that the composition of the relatively proximate upstream agricultural landscape is particularly important for the relationship between precipitation and nitrogen concentrations measured downstream.

5.5 Marginal Effects and Economic Magnitudes

Because the coefficients on triple interactions are difficult to interpret directly, Figure 5 presents the estimated marginal crop–precipitation relationship across the observed distribution of pasture acreage within 50 km upstream. For the row-crop specification, the relevant marginal relationship can be written as

$$\frac{\partial N}{\partial(\text{Row Crop} \times \text{Precipitation})} = \hat{\beta}_{RP} + \hat{\beta}_{RGP}G, \quad (7)$$

where $\hat{\beta}_{RP}$ is the estimated row-crop–precipitation interaction, $\hat{\beta}_{RGP}$ is the estimated row-crop–pasture–precipitation interaction, and G denotes grass/pasture acreage within 50 km upstream. The confidence intervals shown in Figure 5 incorporate the estimated variance and covariance of both interaction coefficients.

The marginal-effect estimates demonstrate that the magnitude of the positive row-crop–precipitation relationship declines considerably across the observed distribution of pasture

acreage. At approximately the 10th percentile of the pasture distribution, corresponding to 343 acres of pasture within 50 km upstream, the estimated marginal effect is 0.0639. At the median, corresponding to approximately 9,455 acres of pasture, the estimated effect falls to 0.0584. At the 90th percentile, where approximately 50,470 acres of pasture are present within 50 km upstream, the estimated marginal effect declines further to 0.0338.

Moving from the 10th to the 90th percentile of observed pasture acreage therefore reduces the estimated magnitude of the row-crop–precipitation relationship by approximately 47 percent. The estimated relationship nevertheless remains positive and statistically distinguishable from zero through the 90th percentile of the pasture distribution. Thus, the estimates are more consistent with attenuation of the row-crop–precipitation relationship than with its complete elimination at pasture levels commonly observed in the sample.

At still higher levels of pasture acreage, the estimated relationship becomes less precisely estimated. The lower bound of the 95-percent confidence interval first crosses zero at approximately 75,000 acres of pasture within 50 km in the row-crop specification. This value lies near the upper tail of the observed pasture distribution and should not be interpreted as a discrete threshold at which pasture eliminates the relationship between row-crop acreage, precipitation, and nitrogen concentration. Rather, it illustrates the declining magnitude and increasing uncertainty of the estimated relationship as pasture acreage becomes large.

The corresponding estimates using only corn acreage produce a similar pattern. The estimated corn–precipitation relationship declines by approximately 48 percent between the 10th and 90th percentiles of the pasture distribution. Because the combined row-crop measure provides a broader characterization of annual row-crop production, Figure 5 reports the row-crop results in the main text, while the corresponding corn estimates are reported in 6

Taken together, the results across Tables 2–5 and Figure 5 reveal a consistent empirical pattern. Recent precipitation is associated with a stronger relationship between upstream corn and soy acreage and downstream nitrogen concentrations; this relationship is concentrated primarily within the nearest 50 km of the upstream network; and greater pasture acreage within that same proximate upstream landscape is associated with a substantial attenuation of the crop–precipitation relationship.

6 Conclusion

Agricultural nonpoint-source pollution presents a persistent challenge for water quality, particularly as precipitation mobilizes nutrients from agricultural landscapes and transports them into surface water. Building on Metaxoglou and Smith (2025), this paper examines how upstream agricultural land use interacts with recent precipitation to shape downstream nitrogen concentrations. In particular, I ask whether replacing annual row-crop production with perennial grass and pasture may reduce the sensitivity of downstream nitrogen concentrations to precipitation.

To examine this question, I combine surface-water nitrogen measurements from USGS mon-

itoring locations with USDA Cropland Data Layer land-use data, PRISM climate data, and the HydroBASINS hydrological network. The resulting estimation sample contains 111,563 nitrogen observations from 4,094 monitoring stations between 2008 and 2023. Monitoring locations are linked to 22,071 unique Level 10 HydroBASINS sub-basins within 300 km upstream. The empirical design explicitly traces agricultural land use through the upstream hydrological network and matches nitrogen observations to crop acreage within the appropriate upstream watersheds and precipitation conditions relevant to the date on which each sample was collected. This makes it possible to examine not only whether agricultural land use is associated with nitrogen concentrations, but how that relationship changes with precipitation and the composition of the upstream agricultural landscape.

Two sets of results provide evidence that substituting grass and pasture for annual row crops may reduce downstream nitrogen concentrations following precipitation. First, corn and soybean acreage exhibit positive and statistically significant interactions with recent precipitation. For corn, this relationship is present over one-, three-, and six-month precipitation windows, while for soybeans it is present over the one- and three-month windows. Grass and pasture acreage, in contrast, does not exhibit a statistically significant positive interaction with precipitation over the one- or three-month windows. Thus, the relationship between agricultural acreage and nitrogen following precipitation differs substantially across land uses: greater precipitation strengthens the relationship between annual row-crop acreage and downstream nitrogen concentrations, while the same pattern is not generally observed for grass and pasture.

Second, the pasture-moderation specifications provide complementary evidence. When row-crop acreage, pasture acreage, and precipitation are allowed to interact, the estimated crop–pasture–precipitation interaction is negative and statistically significant for both corn and combined corn and soy acreage. Greater pasture acreage therefore attenuates the positive relationship between row-crop acreage, precipitation, and downstream nitrogen concentrations. This result is particularly important because it does not rely solely on comparing the separate relationships of row crops and pasture with nitrogen. Instead, it shows that within agricultural landscapes containing row crops, the estimated nitrogen response associated with row crops and precipitation becomes smaller as the amount of pasture in the upstream landscape increases.

Both relationships are strongly spatial in nature. The positive interactions between precipitation and corn and soybean acreage are concentrated within 50 km upstream of the monitoring location. The pasture-moderation effect is similarly strongest within this first 50 km. The same portion of the upstream landscape in which row-crop acreage exhibits the strongest relationship with nitrogen following precipitation is therefore also the portion in which pasture most strongly attenuates that relationship. This spatial correspondence provides additional evidence that the composition of the nearby upstream agricultural landscape matters for downstream water quality.

The marginal-effect estimates illustrate the magnitude of this moderation. In the combined row-crop specification, moving from the 10th to the 90th percentile of pasture acreage within 50 km upstream reduces the estimated marginal row-crop–precipitation relationship by ap-

proximately 47 percent. The estimated relationship remains positive and statistically distinguishable from zero through the 90th percentile of pasture acreage, indicating substantial attenuation rather than complete elimination of the row-crop–precipitation relationship.

Taken together, these findings suggest that replacing some corn and soybean acreage with grass or pasture could reduce the sensitivity of downstream nitrogen concentrations to precipitation. The evidence supporting this interpretation comes from two complementary comparisons: row crops exhibit positive crop–precipitation interactions that are generally absent for pasture, and greater pasture acreage directly attenuates the estimated row-crop–precipitation relationship. The results therefore point toward agricultural land-use composition, rather than row-crop acreage alone, as an important determinant of water-quality responses to precipitation.

This interpretation nevertheless requires an important qualification. The estimates do not directly identify the causal effect of converting a particular acre of corn or soybeans to pasture. Differences across landscapes may reflect additional characteristics correlated with land use, and the analysis does not identify the specific biophysical mechanism responsible for the estimated moderation. Consequently, the results should not be interpreted as an acre-for-acre treatment-effect estimate of land conversion. Rather, they provide evidence consistent with the hypothesis that shifting the composition of agricultural landscapes away from annual row crops and toward perennial grass and pasture would reduce nitrogen responses to precipitation. Directly estimating the effects of observed row-crop-to-pasture transitions would provide a valuable next step for establishing the causal magnitude of such land-use changes.

These findings have potentially important implications for agricultural water-quality policy. Nutrient-management efforts frequently emphasize changes in fertilizer application or management practices within individual row-crop fields. The results here suggest that the composition of the agricultural landscape itself may also matter. Maintaining or expanding perennial grass and pasture within row-crop-dominated upstream landscapes may provide a complementary approach for reducing nitrogen pollution, particularly following precipitation events when the relationship between row-crop production and downstream nitrogen is strongest. More broadly, the results suggest that agricultural production, weather, hydrological proximity, and land-use composition should be considered jointly when evaluating strategies to improve water quality while maintaining productive agricultural landscapes.

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7 Figures

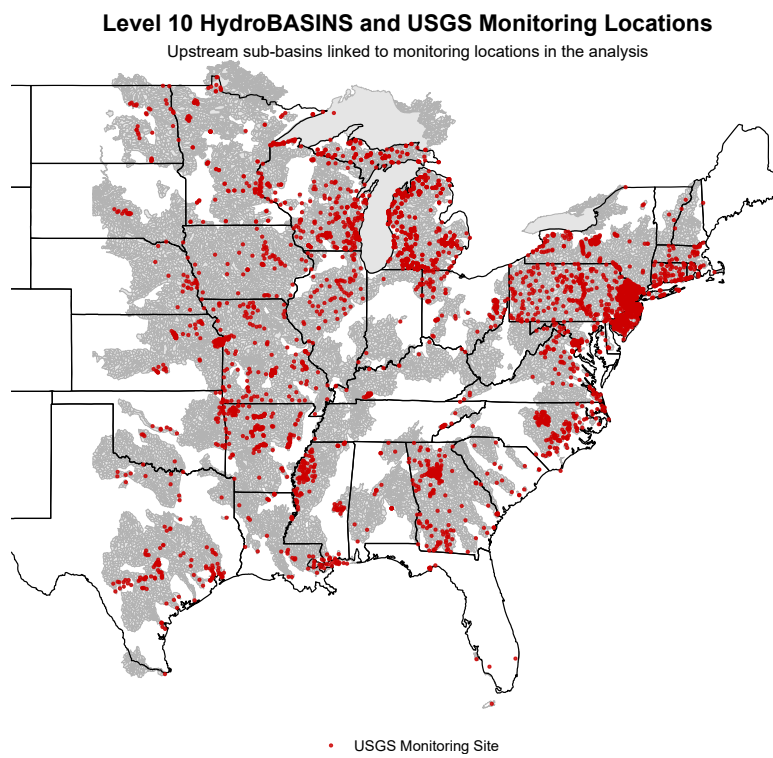


Figure 1: USGS Monitoring Locations and Level 10 HydroBASINS

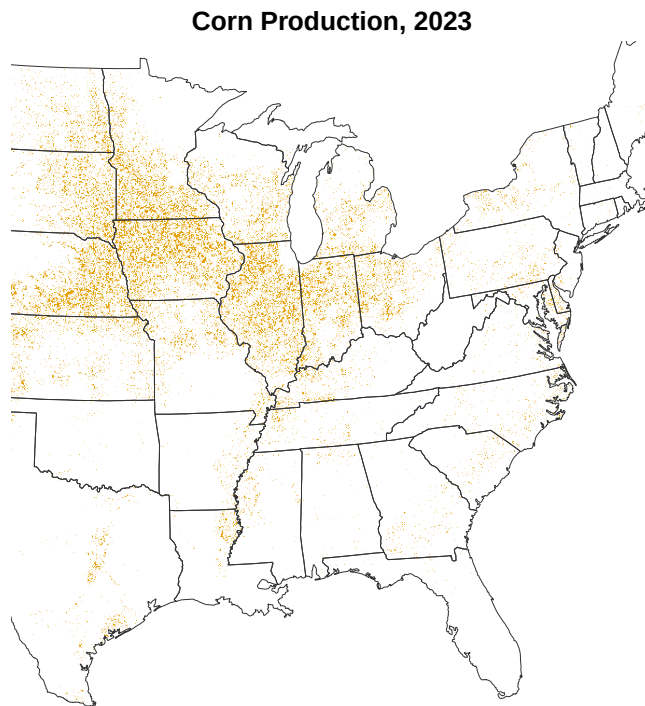


Figure 2: Spatial Distribution of Corn Production, 2023

Soybean Production, 2023

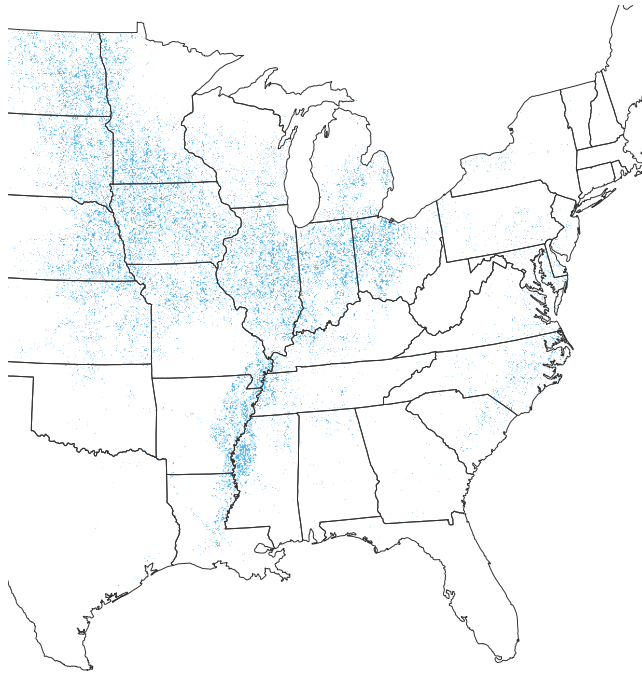


Figure 3: Spatial Distribution of Soybean Production, 2023

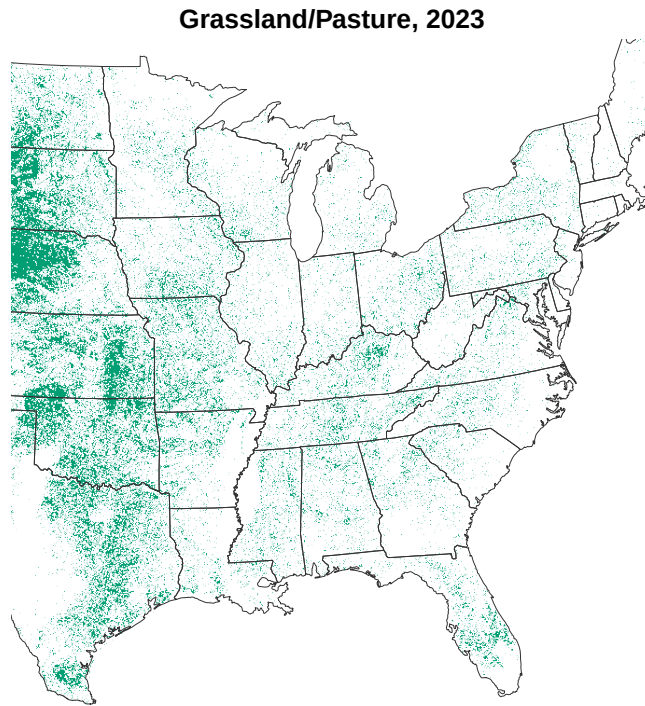


Figure 4: Spatial Distribution of Grassland/Pasture, 2023

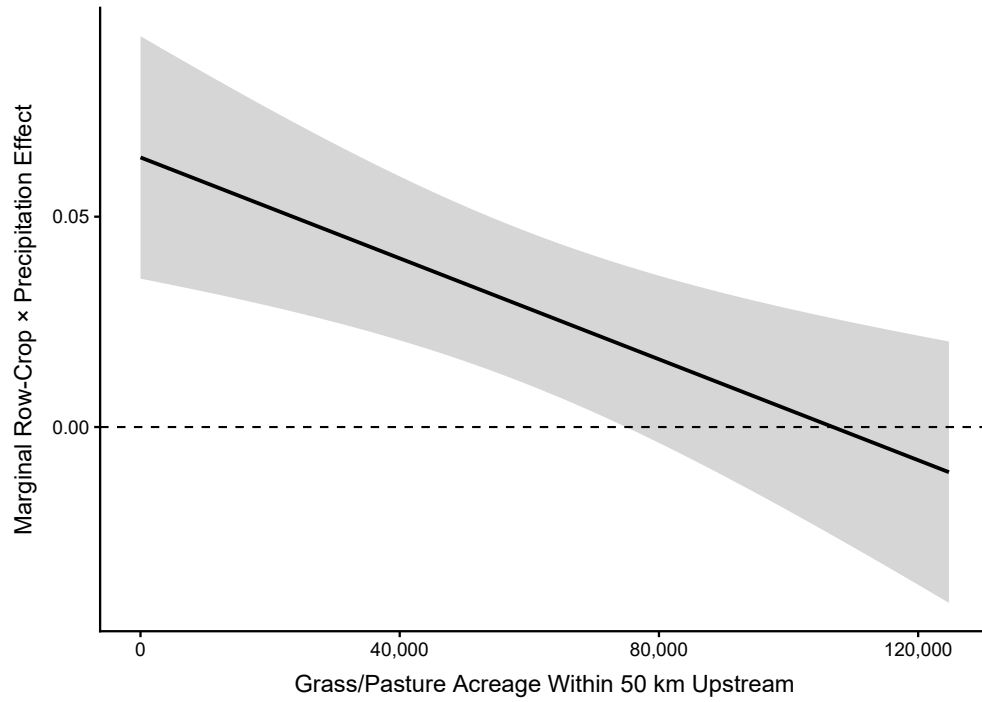


Figure 5: Marginal Row-Crop–Precipitation Relationship by Upstream Pasture Acreage

Notes: The figure reports the estimated marginal relationship between row-crop acreage and precipitation across the observed distribution of grass/pasture acreage within 50 km upstream. The solid line represents the estimated marginal effect and the shaded region reports the 95 percent confidence interval calculated using monitoring-site-clustered standard errors. Grass/pasture acreage is displayed in acres.

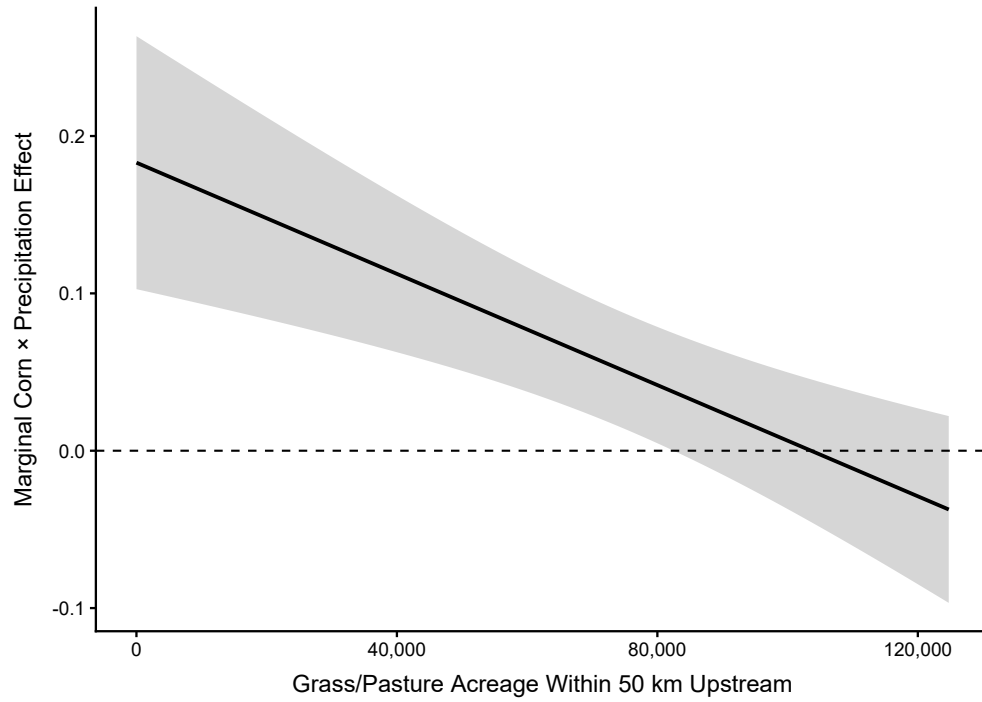


Figure 6: Marginal Corn–Precipitation Relationship by Upstream Pasture Acreage

Notes: The figure reports the estimated marginal relationship between corn acreage and precipitation across the observed distribution of grass/pasture acreage within 50 km upstream. The solid line represents the estimated marginal effect and the shaded region reports the 95 percent confidence interval calculated using monitoring-site-clustered standard errors. Grass/pasture acreage is displayed in acres.

8 Tables

Table 1: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75
<i>Panel A: Upstream Area Within 300 km</i>						
Nitrogen Concentration (mg/L)	115,083	3.00	4.23	0.78	1.50	3.60
Corn Acreage (1,000 acres)	112,668	131.38	418.43	0.62	6.72	43.46
Soybean Acreage (1,000 acres)	112,668	127.19	375.81	0.22	3.11	48.09
Row-Crop Acreage (1,000 acres)	112,668	258.57	781.24	1.12	10.34	103.03
Grass/Pasture Acreage (1,000 acres)	112,668	146.69	371.61	2.99	14.36	120.74
1-Month Precipitation (mm)	115,083	104.23	59.02	63.42	94.67	133.62
3-Month Mean Precipitation (mm)	115,083	99.95	40.33	72.18	94.85	122.55
6-Month Mean Precipitation (mm)	115,083	96.50	31.11	75.22	94.34	115.01
Annual Mean Temperature (C)	115,083	11.68	3.73	8.86	11.15	14.23
January Minimum Temperature (C)	115,083	-6.51	5.53	-10.29	-6.43	-2.82
August Maximum Temperature (C)	115,083	28.63	2.73	26.66	28.21	30.38
<i>Panel B: Upstream Area Within 0–50 km</i>						
Row-Crop Acreage (1,000 acres)	112,668	36.48	72.87	0.65	7.50	35.28
Grass/Pasture Acreage (1,000 acres)	112,668	19.35	26.94	2.53	9.46	25.26
1-Month Precipitation (mm)	115,083	104.70	60.44	62.86	94.55	134.03

Notes: Descriptive statistics are calculated across nitrogen observations satisfying the sample restrictions described in the text. Nitrogen concentration is measured in mg/L. Crop and pasture acreage are reported in thousands of acres within the indicated upstream distance. Precipitation is measured in millimeters and temperature in degrees Celsius. Crop acreage is unavailable for observations assigned to crop years outside the coverage of the Cropland Data Layer, so the number of observations differs across variables.

Table 2: Upstream Crop Acreage, Precipitation, and Nitrogen Concentration

Precipitation Window Model	Corn			Soybean			Grass/Pasture		
	1 Month (1)	3 Month (2)	6 Month (3)	1 Month (4)	3 Month (5)	6 Month (6)	1 Month (7)	3 Month (8)	6 Month (9)
Corn Acreage	0.6714 (0.5605)	0.5442 (0.5556)	0.5000 (0.5757)						
Corn Acreage $\times$ Precipitation	0.0048*** (0.0010)	0.0066*** (0.0013)	0.0052*** (0.0019)						
Soybean Acreage				-0.4535 (0.3890)	-0.4890 (0.4326)	-0.1697 (0.5118)			
Soybean Acreage $\times$ Precipitation				0.0054*** (0.0011)	0.0062*** (0.0015)	0.0029 (0.0026)			
Grass/Pasture Acreage							-0.1189 (0.2278)	-0.2460 (0.2968)	-0.5015 (0.3070)
Grass/Pasture Acreage $\times$ Precipitation							0.0015 (0.0012)	0.0030 (0.0023)	0.0052** (0.0022)
Precipitation	0.0010 (0.0014)	-0.0057 (0.0041)	-0.0288*** (0.0057)	0.0010 (0.0015)	-0.0052 (0.0041)	-0.0274*** (0.0057)	0.0018 (0.0015)	-0.0039 (0.0041)	-0.0282*** (0.0057)
Site Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Temperature Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Precipitation Squared	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	111,563	111,563	111,563	111,563	111,563	111,563	111,563	111,563	111,563
R ²	0.62543	0.62551	0.62782	0.62541	0.62531	0.62769	0.62486	0.62505	0.62780
Within R ²	0.00281	0.00301	0.00916	0.00275	0.00247	0.00880	0.00127	0.00178	0.00911

Notes: The dependent variable is nitrogen concentration (mg/L). Crop acreage is measured in millions of acres within 300 km upstream of the monitoring location. Precipitation is the area-weighted mean precipitation over the indicated window ending in the observation month. All specifications include a quadratic precipitation term, annual mean temperature, January minimum temperature, August maximum temperature, monitoring-site fixed effects, and observation-year fixed effects. Standard errors clustered by monitoring site are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Spatial Distribution of the Crop–Precipitation Relationship

Dependent Variable: Model:	Nitrogen Concentration (Corn) (Soy)	
<i>Variables</i>		
0–50 km	0.1229*** (0.0274)	0.0703*** (0.0154)
50–100 km	-0.0026 (0.0133)	-0.0018 (0.0094)
100–150 km	-0.0092 (0.0105)	-0.0097 (0.0099)
150–200 km	0.0033 (0.0109)	0.0077 (0.0094)
200–250 km	0.0081 (0.0104)	0.0069 (0.0092)
250–300 km	-0.0085 (0.0078)	-0.0048 (0.0066)
<i>Fixed-effects</i>		
site_id	Yes	Yes
obs_year	Yes	Yes
<i>Fit statistics</i>		
Observations	111,563	111,563
R ²	0.62750	0.62677
Within R ²	0.00830	0.00637

Clustered (site_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Reported coefficients are crop acreage $\times$ one-month precipitation interactions for mutually exclusive upstream distance rings.

Crop acreage is measured in millions of acres and precipitation in millimeters.

All six distance rings enter each specification simultaneously.

Each specification includes the corresponding crop acreage, precipitation, quadratic precipitation terms, annual mean temperature, January minimum temperature, August maximum temperature, monitoring-site fixed effects, and observation-year fixed effects.

Standard errors are clustered by monitoring site.

Table 4: Pasture Moderation of the Crop–Precipitation Relationship

Dependent Variable: Model:	Nitrogen Concentration (Corn) (Row Crops)	
<i>Variables</i>		
Corn Acreage	0.4658 (0.6720)	
Grass/Pasture Acreage	0.2888 (0.3458)	0.2044 (0.3529)
Precipitation	0.0008 (0.0015)	0.0008 (0.0015)
Corn Acreage $\times$ Precipitation	0.0101*** (0.0020)	
Grass/Pasture Acreage $\times$ Precipitation	0.0001 (0.0018)	-0.0001 (0.0018)
Corn Acreage $\times$ Grass/Pasture Acreage	0.2606 (0.1883)	
Corn Acreage $\times$ Grass/Pasture Acreage $\times$ Precipitation	-0.0048*** (0.0017)	
Row-Crop Acreage		0.1122 (0.4593)
Row-Crop Acreage $\times$ Precipitation		0.0049*** (0.0010)
Row-Crop Acreage $\times$ Grass/Pasture Acreage		0.1552 (0.1065)
Row-Crop Acreage $\times$ Grass/Pasture Acreage $\times$ Precipitation		-0.0021** (0.0009)
<i>Fixed-effects</i>		
site_id	Yes	Yes
obs_year	Yes	Yes
<i>Fit statistics</i>		
Observations	111,563	111,563
R ²	0.62570	0.62565
Within R ²	0.00353	0.00339

Clustered (site_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent variable is nitrogen concentration (mg/L).

Row-crop acreage is defined as the sum of corn and soybean acreage.

Crop and grass/pasture acreage are measured in millions of acres within 300 km upstream.

Precipitation is mean precipitation during the observation month.

All specifications include all lower-order terms required by the triple interaction, a quadratic precipitation term, temperature controls, monitoring-site fixed effects, and observation-year fixed effects.

Standard errors are clustered by monitoring site.

Table 5: Spatial Heterogeneity in Pasture Moderation

Dependent Variable: Model:	Nitrogen (Corn)	Concentration (Row Crops)
<i>Variables</i>		
0–50 km	-1.767*** (0.4832)	-0.5998*** (0.1948)
50–100 km	0.1716 (0.1185)	0.1065** (0.0463)
100–150 km	-0.0177 (0.0735)	-0.0032 (0.0357)
150–200 km	-0.0026 (0.0446)	0.0086 (0.0242)
200–250 km	-0.0269 (0.0375)	-0.0173 (0.0201)
250–300 km	0.0226 (0.0233)	0.0120 (0.0123)
<i>Fixed-effects</i>		
site_id	Yes	Yes
obs_year	Yes	Yes
<i>Fit statistics</i>		
Observations	111,563	111,563
R ²	0.62809	0.62767
Within R ²	0.00988	0.00875

Clustered (site_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Reported coefficients are crop acreage $\times$ grass/pasture acreage $\times$ one-month precipitation interactions for mutually exclusive upstream distance rings.

Row-crop acreage is defined as the sum of corn and soybean acreage.

All six distance rings enter each specification simultaneously.

All required lower-order interaction terms, quadratic precipitation terms, temperature controls, monitoring-site fixed effects, and observation-year fixed effects are included but not reported.

Standard errors are clustered by monitoring site.